

Week 1 Introduction



Topic Introduction

Week 1 Lesson Video

Run time: 19:32

Watch this video for a comprehensive introduction to this week's learning content. You can follow along in Blackboard as you watch, with the video organized into the following chapters:

- Lesson 1.1 Deep Learning, Neural Networks, and Artificial Neurons
- Lesson 1.2 Lesson: From a Single Neuron to a Neural Network (06:44)
- Lesson 1.3 Regression and Classification with Neural Networks (09:54)



Required Resources

The following resources are required for your learning this week. Make sure you review everything linked below, as you may be tested on the concepts in an upcoming Knowledge Check or Homework.

Reading | Géron, A. (2023). Chapter 10: Introduction to artificial neural networks. In Hands-on machine learning with scikit-learn, Keras and TensorFlow: Concepts, tools, and techniques to build intelligent systems

(https://bu.primo.exlibrisgroup.com/permalink/01BOSU_INST/u1o13f/alma99209313482001161) (299-308). O'Reilly Media, Inc.

- Read pages 299 to 308 for a concise treatment of the history and current trends in artificial neural networks and how neurons (which the author calls perceptrons) work. Well worth reading for deep context.

Video | Sanderson, G. (2017, October 5). *But what is a neural network?* | *Deep learning chapter 1* (https://www.youtube.com/watch?v=aircAruvnKk&list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi&index=1&t=8s) [Video]. YouTube.

- This is the first in a series of hugely influential video tutorials by the brilliant Grant Sanderson on his **3Blue1Brown** YouTube channel. Essential viewing!



Key Terms

- **Deep Learning:** A subset of machine learning that uses artificial neural networks with many layers to model complex patterns in data.
- **Artificial Neural Network (ANN):** A computational model inspired by biological neural networks consisting of layers of interconnected artificial neurons.
- **Neuron:** A basic computational unit in a neural network that calculates the weighted sum of its inputs, followed by an activation function.
- **Input Layer:** The initial set of raw input values to the network; not composed of neurons.
- **Output Layer:** Final layer that produces the network's prediction or classification result.
- **Hidden Layer:** Intermediate layers of neurons that apply weights and activation functions to transform inputs to outputs.
- **Activation Function:** A function applied to the output of a neuron to introduce non-linearity, enabling the network to learn complex patterns (e.g., sigmoid, ReLU, tanh).
- **Bias Term:** A constant added to the neuron's computation, often implemented as an additional weight.
- **Loss Function:** A mathematical function that measures the error between the predicted and actual targets
- **Parameters:** The weights and biases in the network that are learned during training.
- **Hyperparameters:** Design choices such as the shape of the network, the activation functions, and training settings.

- **Regression (in Neural Networks):** A task where the network outputs a floating-point value, usually by omitting the non-linear activation in the output neuron.
- **Binary Classification (in Neural Networks):** A task where the network assigns an input to the probability that it belongs to a class using a sigmoid in the output layer.
- **Multi-Label Classification (in Neural Networks):** A task where the network predicts an independent sigmoid-based probability for each class, so an input can be associated with multiple classes at once.